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Roll No.

220921

2nd Sem / Electrical
Subject : Electrical Networks

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The Norton equivalent circuit consists of: (CO1)
- a) A voltage source in series with a resistor
 - b) A current source in parallel with a resistor
 - c) A voltage source in parallel with a resistor
 - d) A current source in series with a resistor.
- Q.2 A network that contains any source of energy is called: (CO1)
- a) Active network b) Passive network
 - c) Linear network d) Non-linear network
- Q.3 Which terms refers to the maximum value reached by an AC waveform during one cycle? (CO2)
- a) Amplitude b) Time period
 - c) Frequency d) Instantaneous value

- Q.11 What is mathematical formula for capacitive reactance? (CO3)
- Q.12 What is unbalanced load in polyphase system? (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 List various Causes of low power factor. (CO2)
- Q.14 Derive an equation for RMS value of an alternating quantity? (CO2)
- Q.15 With the help neat sketch explain effects of AC voltage applied to pure Resistor? (CO3)
- Q.16 Explain parallel resonance with suitable diagram? (CO3)
- Q.17 Compare star and Delta connection with suitable diagram? (CO4)
- Q.18 Explain nodal analysis with suitable example? (CO1)
- Q.19 List differences between Linear and Non-linear circuit? (CO1)
- Q.20 Define: Q factor, Average Value of AC (CO2)
- Q.21 Explain the concept of phase and phase difference with suitable example? (CO4)

Q.22 Explain Thevenin Theorem with suitable diagram?
(CO1)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Illustrate the R-C series circuit using an appropriate phase diagram. Additionally describe the concepts of impedance and power triangle for this circuit, accompanied by suitable diagrams. (CO3)

Q.24 Explain in detail, with appropriate diagrams, how a circuit is analyzed using the Norton Theorem?
(CO1)

Q.25 Write a note on : (CO1)

- a) Maximum Power Transfer Theorem
- b) Harmonics in AC